



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Division: A1

ASSIGNMENT No: 5

TITLE: Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Theory

ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

Exercise

1. Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks

Source code:.

```
1  import java.util.ArrayList;
2  import java.util.Scanner;
3
4  public class assignment5_TodoList {
    Run | Debug | Run main | Debug main
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7          ArrayList<String> tasks = new ArrayList<>();
8
9          while (true) {
10             System.out.println(x: "\n===== TO-DO LIST =====");
11             System.out.println(x: "1. Add Task");
12             System.out.println(x: "2. Remove Task");
13             System.out.println(x: "3. View Tasks");
14             System.out.println(x: "4. Exit");
15             System.out.print(s: "Enter your choice: ");
16
17             int choice = sc.nextInt();
18             sc.nextLine();
19
20             if (choice == 1) {
21                 System.out.print(s: "Enter task: ");
22                 String task = sc.nextLine();
23                 tasks.add(task);
24                 System.out.println(x: "Task added successfully.");
25
26             } else if (choice == 2) {
27                 if (tasks.isEmpty()) {
28                     System.out.println(x: "No tasks available.");
29                 } else {
30                     System.out.print(s: "Enter task number to remove: ");
31                     int index = sc.nextInt();
32
33                     if (index >= 1 && index <= tasks.size()) {
34                         tasks.remove(index - 1);
35                         System.out.println(x: "Task removed successfully.");
36                     } else {
37                         System.out.println(x: "Invalid task number.");
38                     }
39                 }
40             }
41         }
42     }
43 }
```

```

39     }
40
41     } else if (choice == 3) {
42         if (tasks.isEmpty()) {
43             System.out.println(x: "No tasks available.");
44         } else {
45             StringBuffer sb = new StringBuffer();
46             sb.append(str: "\n----- TO-DO TASKS ----- \n");
47
48             for (int i = 0; i < tasks.size(); i++) {
49                 sb.append(i + 1)
50                   .append(str: ". ")
51                   .append(tasks.get(i))
52                   .append(str: "\n");
53             }
54
55             System.out.println(sb);
56         }
57
58     } else if (choice == 4) {
59         System.out.println(x: "Exiting...");
60         break;
61
62     } else {
63         System.out.println(x: "Invalid choice.");
64     }
65 }
66
67 sc.close();
68 }
69 }

```

Output:

```

C:\Users\Aashray\Documents\University\3rd sem\JAVA>java assignment5_TodoList.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8

===== TO-DO LIST =====
1. Add Task
2. Remove Task
3. View Tasks
4. Exit
Enter your choice: 1
Enter task: complete java assignment
Task added successfully.

```

```
===== TO-DO LIST =====
1. Add Task
2. Remove Task
3. View Tasks
4. Exit
Enter your choice: 3

----- TO-DO TASKS -----
1. complete java assignment
```

```
===== TO-DO LIST =====
1. Add Task
2. Remove Task
3. View Tasks
4. Exit
Enter your choice: 2
Enter task number to remove: 1
Task removed successfully.
```

```
===== TO-DO LIST =====
1. Add Task
2. Remove Task
3. View Tasks
4. Exit
Enter your choice: 4
Exiting...
1. Add Task
2. Remove Task
3. View Tasks
4. Exit
Enter your choice: 4
Exiting...
```

2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

Source Code

```
1  import java.util.ArrayList;
2  import java.util.Scanner;
3
4  public class assignment5_CourseRegistration {
5      Run | Debug | Run main | Debug main
6      public static void main(String[] args) {
7          Scanner sc = new Scanner(System.in);
8          ArrayList<String> courses = new ArrayList<>();
9
10         while (true) {
11             System.out.println(x: "\n===== COURSE REGISTRATION =====");
12             System.out.println(x: "1. Register Course");
13             System.out.println(x: "2. Remove Course");
14             System.out.println(x: "3. View Registered Courses");
15             System.out.println(x: "4. Exit");
16             System.out.print(s: "Enter your choice: ");
17
18             int choice = sc.nextInt();
19             sc.nextLine();
20
21             if (choice == 1) {
22                 System.out.print(s: "Enter course name: ");
23                 String course = sc.nextLine();
24                 courses.add(course);
25                 System.out.println(x: "Course registered successfully.");
26             } else if (choice == 2) {
27                 if (courses.isEmpty()) {
28                     System.out.println(x: "No registered courses.");
29                 } else {
30                     System.out.print(s: "Enter course number to remove: ");
31                     int index = sc.nextInt();
32
33                     if (index >= 1 && index <= courses.size()) {
34                         courses.remove(index - 1);
35                         System.out.println(x: "Course removed successfully.");
36                     } else {
37                         System.out.println(x: "Invalid course number.");
38                     }
39                 }
40             }
41         }
42     }
43 }
```

:

```
39     }
40
41     } else if (choice == 3) {
42         if (courses.isEmpty()) {
43             System.out.println(x: "No registered courses.");
44         } else {
45             StringBuffer sb = new StringBuffer();
46             sb.append(str: "\n----- REGISTERED COURSES ----- \n");
47
48             for (int i = 0; i < courses.size(); i++) {
49                 sb.append(i + 1)
50                   .append(str: ". ")
51                   .append(courses.get(i))
52                   .append(str: "\n");
53             }
54
55             System.out.println(sb);
56         }
57
58     } else if (choice == 4) {
59         System.out.println(x: "Exiting...");
60         break;
61
62     } else {
63         System.out.println(x: "Invalid choice.");
64     }
65 }
66
67 sc.close();
68 }
69 }
```

Output:

```
C:\Users\Aashray\Documents\University\3rd sem\JAVA>java assignment5_CourseRegistration.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8

===== COURSE REGISTRATION =====
1. Register Course
2. Remove Course
3. View Registered Courses
4. Exit
Enter your choice: 1
Enter course name: JAVA
Course registered successfully.
```

```
===== COURSE REGISTRATION =====
1. Register Course
2. Remove Course
3. View Registered Courses
4. Exit
Enter your choice: 3

----- REGISTERED COURSES -----
1. JAVA
```

```
===== COURSE REGISTRATION =====
1. Register Course
2. Remove Course
3. View Registered Courses
4. Exit
Enter your choice: 4
Exiting...
Enter course number to remove: 1
Course removed successfully.
```

```
===== COURSE REGISTRATION =====
1. Register Course
2. Remove Course
3. View Registered Courses
4. Exit
Enter your choice: 2
Enter course number to remove: 1
Course removed successfully.
```

```
===== COURSE REGISTRATION =====
1. Register Course
2. Remove Course
3. View Registered Courses
4. Exit
Enter your choice: 4
Exiting...
```